AI-DRIVEN CYBERSECURITY MONITORING AGENT

EcoDrive AI Agent

Autonomous Vehicle Monitoring & Intelligent Threat Detection System

Team: AI Cyber Sentinels   Track: AI-Driven Cybersecurity

Python | Jupyter | Scikit-Learn | SQLite | Pandas & NumPy

The Challenge: Outpaced Vehicle Monitoring

The Core Issue

Modern vehicles operate as hyper-connected IoT devices, continuously streaming multi-gigabyte telemetry packets. However, legacy monitoring remains primitive and retroactive.

- **Delayed Anomaly Detection:** Infrequent polling means failures are realized only after damage is done.
- **Zero Autonomy:** Over-reliance on manual garage diagnostics.
- **Excessive Overhead:** Severe cost spikes driven by unexpected, unpredicted structural breakdowns.

OUR MISSION GOAL

Continuous Autonomous Defense

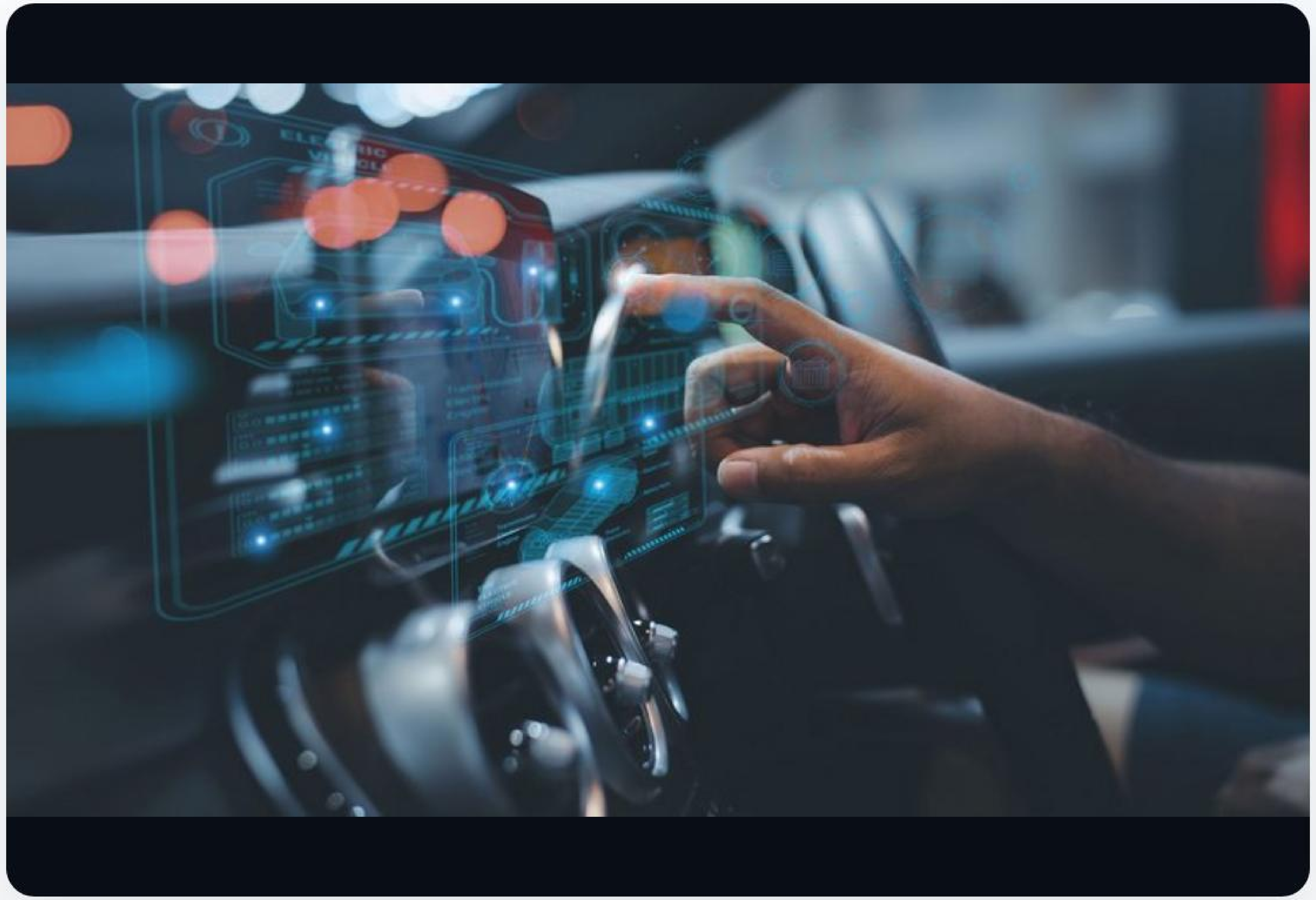
Engineered to construct a proactive, completely automated AI Agent workflow capable of processing incoming OBD-II telemetry in real-time, accurately categorizing threats, classifying risk, and executing containment protocols autonomously.

The Solution: EcoDrive AI Agent

Unified Multi-Agent Security Engine

Our solution harnesses deep analytical monitoring and ML risk calculation, deploying decentralized agents to ensure vehicle structural integrity.

- **Real-time Telemetry Parsing:** Steady streams of OBD-II parameters.
- **ML Risk Classification:** Predicts faults before physical manifestation.
- **Autonomous Alert Dispatch:** Instant warning generation.
- **Diagnostic Report Output:** Generates actionable fleet logs.



Decentralized Multi-Agent Architecture

Monitoring

Ingests OBD-II telemetry and cleans stream data.

Feature Eng.

Calculates rolling parameters and metrics.

Detection

Random Forest detects telemetry anomalies.

Decision

Assesses hazard severity and risks.

Response

Triggers immediate critical fail-safes.

Report

Creates detailed maintenance plans.

End-to-End Operational Workflow

1. Ingestion

OBD-II vehicle sensors generate continuous telemetry streams (coolant temperatures, engine load percentages, fuel efficiency rates, RPM anomalies).

2. Processing

Data streams undergo engineering before being evaluated by the **Random Forest Anomaly model** to quantify severity thresholds.

3. Action

The Decision Agent reviews threat levels, instantly prompting local operator dashboards and cataloging automated reports directly in SQLite databases.

System Technology Blueprint

Layer Category	Technology Selected	Architectural Intent & Role
Language Core	Python, Jupyter Notebook	High-velocity algorithmic development and script deployment.
Data Processing	Pandas, NumPy	Telemetry ingestion, matrix operations, and feature manipulation.
Machine Learning	Scikit-learn (Random Forest)	Supervised model training, predictions, and multi-class classifications.
Infrastructure DB	SQLite	Relational logging of telemetry packets and diagnostic reports.
Visualization	Matplotlib, Seaborn	Rendering local confusion matrices and feature relevance curves.

Features Implemented & Completed

Core Data & Sim Engine

- **Synthetic Telemetry Gen:** Engineered realistic OBD-II sensory behavior data.
- **Feature Engineering:** Automated rolling averages and variance indicators.
- **Local SQLite Storage:** Instant local structured querying of logs.

Intelligent Agent AI Stack

- **Random Forest Classifier:** High accuracy risk categorization.
- **Autonomous Alert Dispatch:** Rule-based urgent notification triggers.
- **Comprehensive Diagnosis:** Auto-generated system condition logs.

Performance Results & Model Accuracy

MODEL ACCURACY SCORE

97.4%

Verified with a 20% validation split from real-time synthetic data.

96.8%
Precision

97.1%
Recall Rate

Benchmark Comparison

Evaluating prediction models on vehicular data streams.

Massive Cloud Scale & Future Scope

Enterprise Grade Fleet Monitoring

EcoDrive AI Agent isn't just a prototype; it's designed to act as an edge-ready solution to oversee multiple fleets concurrently with minimal network overhead.

- **Multi-Vehicle Scalability:** Supports thousands of concurrent units.
- **Edge AI Integration:** Moving RF models straight into edge compute modules.
- **CAN Bus Telemetry:** Expanding into hardware diagnostic buses.

Tangible Market & Fleet Impact

Reduced Fatalities

Mitigates critical brake and engine failures through early sensor degradation tracking.

30% Cost Reduction

Enables proactive maintenance scheduling, bypassing emergency vehicle retrieval costs.

Extended Lifespans

Prevents prolonged operations under elevated temperatures or excessive stress.

THANK YOU FOR YOUR TIME

EcoDrive AI Agent

The Future of Proactive, Autonomous Vehicle Health & Diagnostics

Development Track

AI-Driven Cybersecurity

Project Repository

EcoDrive System Suite